

Introduction

Aircraft efficiency is a major focus of modern aerospace engineering, with **winglets** playing a key role in reducing wingtip vortices, drag, and **fuel consumption**. While **fixed-shape winglets** can improve efficiency by **4–6%**, and **split scimitar winglets** add another **1–3%**, they are limited because they are optimized for only one flight condition, usually cruising conditions. Airlines spent approximately **\$291 billion** on jet fuel in 2024 (IATA, 2024). Even a small improvement in efficiency can result in **billions of dollars in savings** and **reduced emissions**.

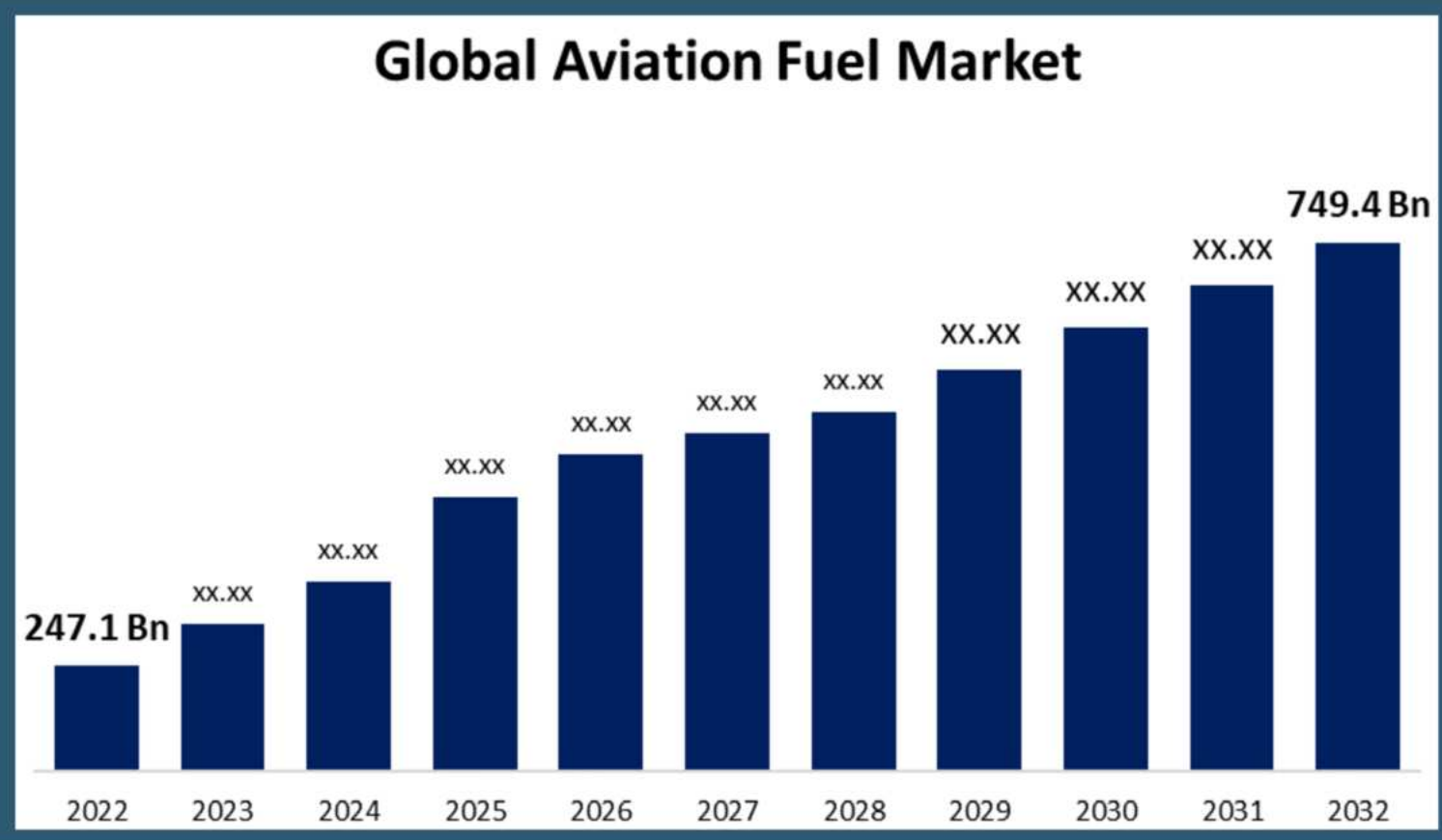


Figure 1: Fuel Cost (Spherical Insights, 2023)

Adapative winglets address the limitation of fixed winglets by investigating a **morphing winglet** that can change shape during flight. Using **shape memory alloys (SMAs)**, specifically **nitinol**, the winglet transitions between a **blended wingtip** for takeoff and a **split scimitar shape** for cruising. A **3D-printed prototype** was designed and tested in a **wind tunnel** to measure changes in **lift-to-drag ratio**. Testing has determined whether **adaptive winglets** can improve **aerodynamic efficiency** and support more sustainable aircraft design.

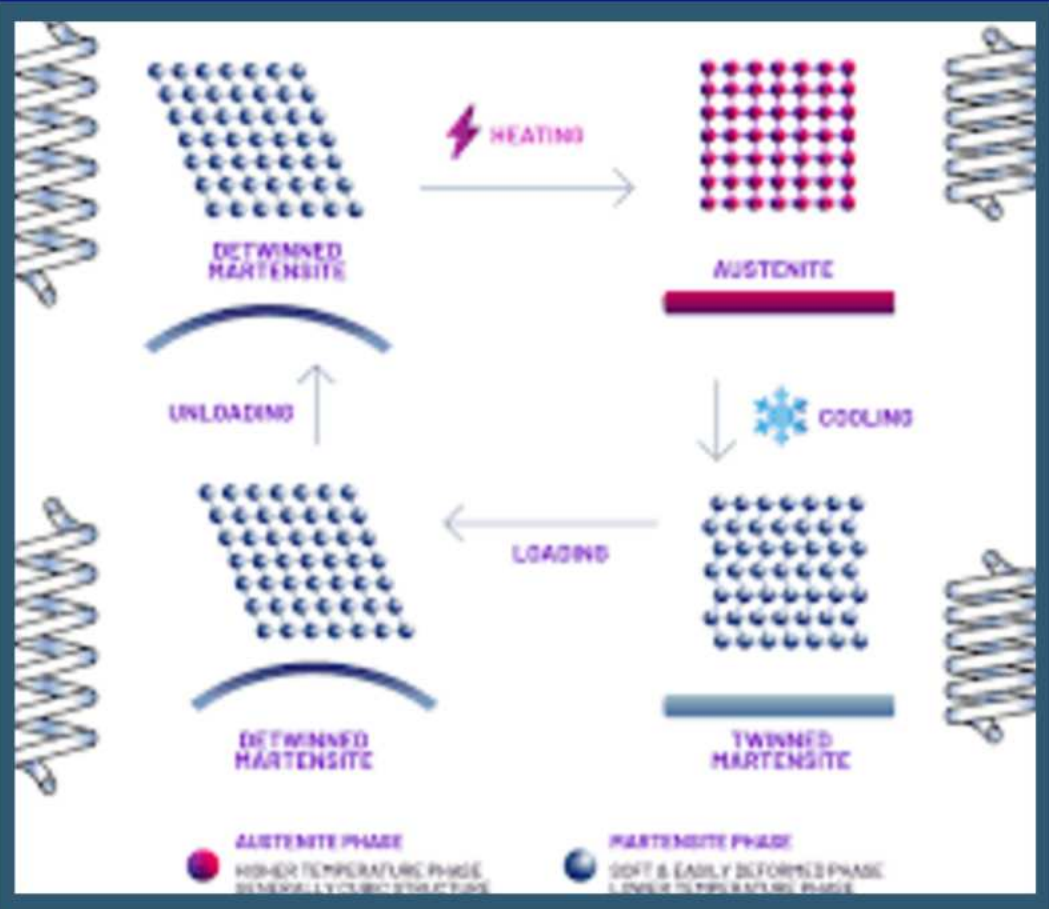


Figure 2: Nitinol Activation (Cambridge Mechatronics, 2023)

Purpose

To design, build, and test a **shape-changing winglet** using **shape memory alloys (SMAs)**. By comparing **lift-to-drag ratios** during different configurations and different speeds through **wind tunnel testing**., The wind tunnel will evaluate whether morphing winglets can improve aircraft efficiency and **reduce fuel consumption**.

Implementing Adaptive Winglets Utilizing Shape Memory Alloys

Methods

Modeling And Fabrication

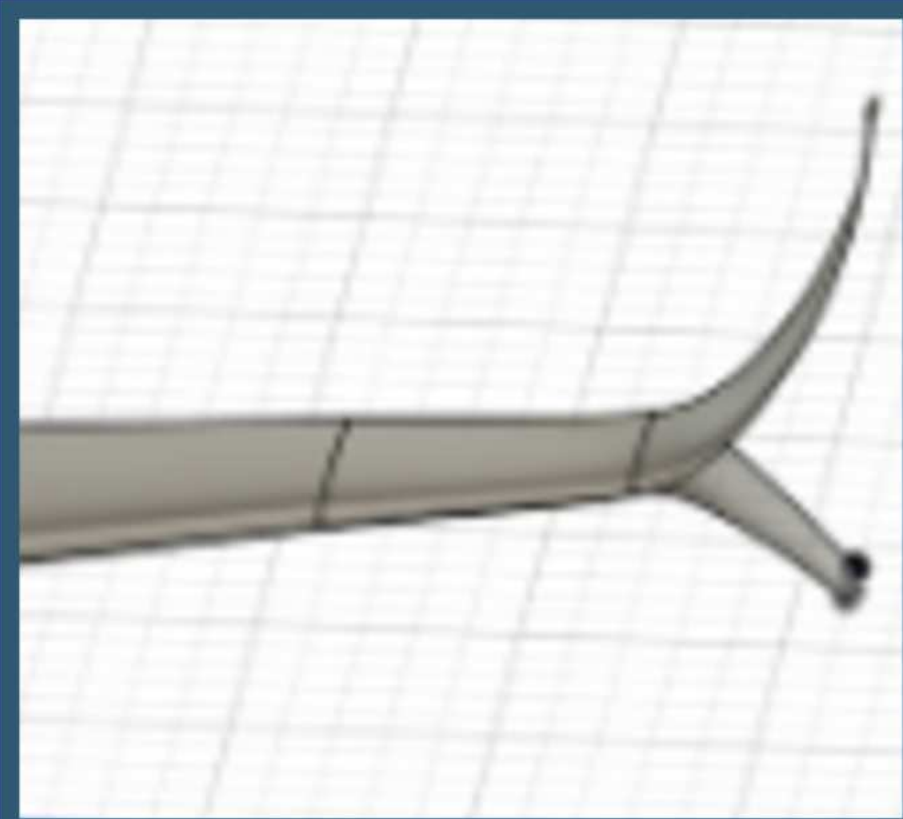


Figure 3: CAD Prototype (CAD Model Created By Finalist Using Fusion 360, 2026)



Figure 4: Split Scimitar Wing (Photo Taken by Finalist, 2026)



Figure 5: Blended Wingtip (Photo Taken by Finalist, 2026)

- Designed a 3D wing and winglet system in Fusion 360 with blended and split scimitar configurations.
- 3D printed the wing and built the ventral strake using trained Nitinol (SMA) wire, secured with heat-resistant epoxy.



Figure 6: Nitinol Ventral Strake (Photo Taken by Finalist, 2026)



Figure 7: Ventral Strake With Paper (Photo Taken by Finalist, 2026)

Wind Tunnel Test

- Tested both shapes at multiple airspeeds and angles, measured lift and drag, and calculated L/D ratios.
- Compared results to evaluate performance and reliability.

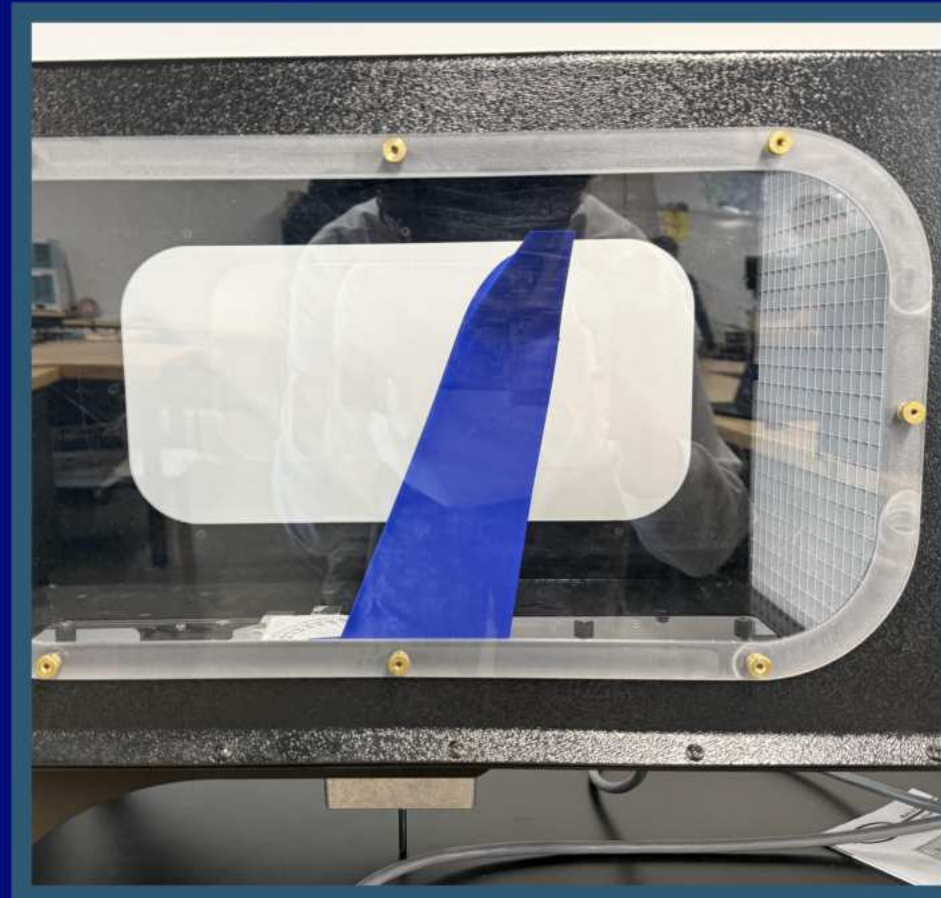


Figure 8: Wind Tunnel Setup (Photo Taken by Finalist, 2026)

Results

Blended Wingtip					
Speed (ft/min)	Angle	L/D	Angle	L/D	
1000	3	7	15	7	
1500	3	9	15	7.2	
2000	3	15.5	15	7.625	
2500	3	22.5	15	8.09	
3000	3	22.33	15	8.18	
3500	3	22	15	8.79	
4000	3	22	15	8.73	
Smooth Split Scimitar					
Speed (ft/min)	Angle	L/D	Angle	L/D	
1000	3	12	15	5.33	
1500	3	13	15	5.43	
2000	3	21	15	5.46	
2500	3	27	15	6.56	
3000	3	26.67	15	7.05	
3500	3	34	15	7.44	
4000	3	34.5	15	7.49	

Figure 9: Lift to Drag Ratio Comparison Table (Data Table Created by Finalist using Google Sheets, 2026)



Figure 10: Takeoff Graph (Graph Created by Finalist Using Google Sheets, 2026)

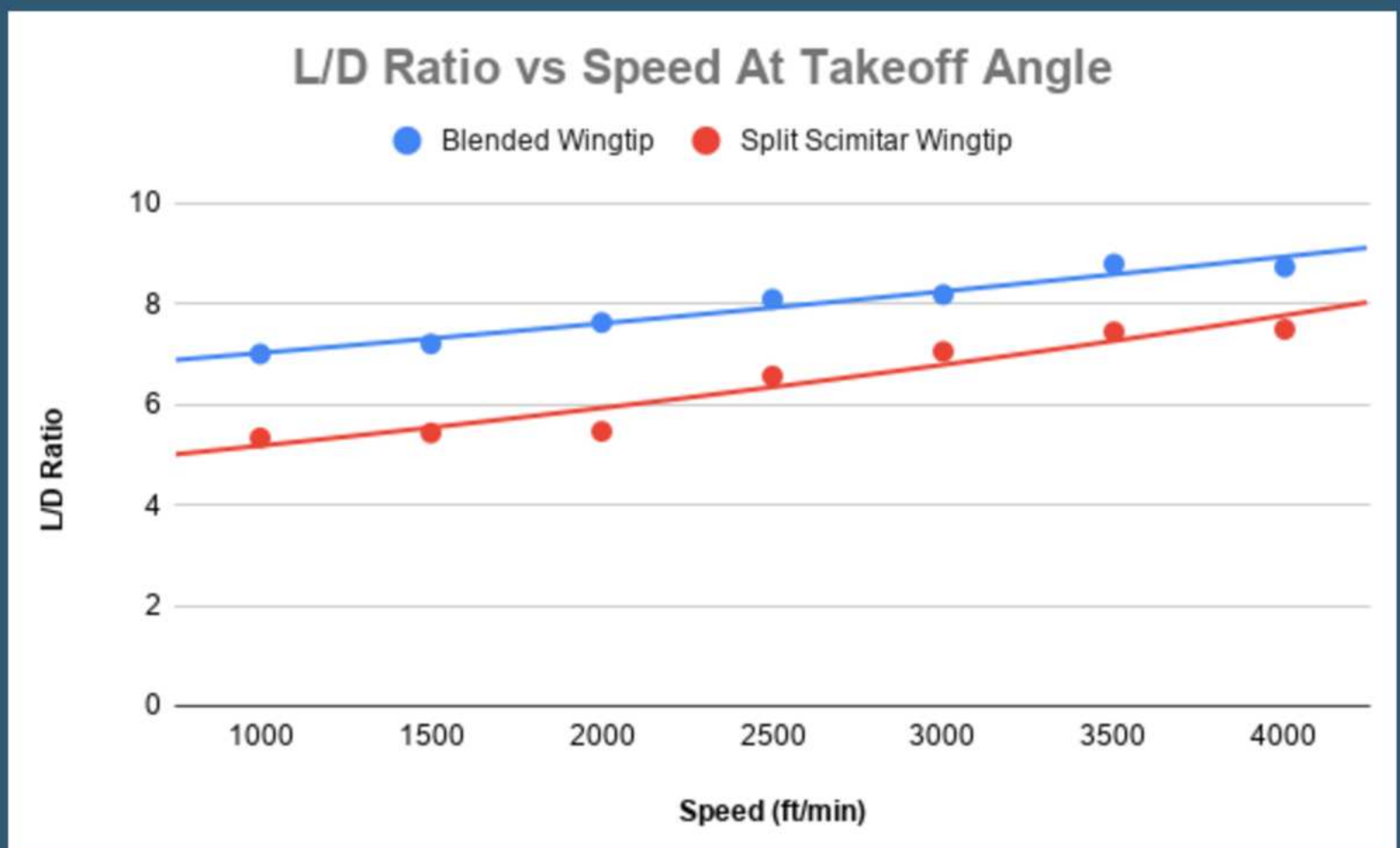


Figure 11: Cruising Graph (Graph Created by Finalist Using Google Sheets, 2026)

Conclusions

At the takeoff angle, the **blended wingtip** had the **higher lift-to-drag ratio**, making it more efficient in low-speed, high-lift conditions. At the **cruising angle**, the **split scimitar wingtip** produced the **higher L/D ratio**, especially as wind tunnel speed increased. As velocity increased, the split scimitar shape became more efficient, which reflects real flight conditions as aircraft accelerate toward cruise. These results show that each winglet performs best at different phases of flight and support that **mid-flight shape change improves aerodynamic efficiency**.

Extensions



Figure 12: Electrical Current Activation (Photo Taken by Finalist, 2026)

One important extension of this project was **activating the Nitinol (SMA) using an electrical current** instead of external heating. By passing a controlled current through the wire, the SMA heated successfully changed the winglet from the blended configuration to the split scimitar configuration. This makes the system **more realistic for real aircraft applications**, where electrical systems are already available onboard. In the future, continued testing with different **ventral strake angles** to determine the most aerodynamically optimal configuration will occur. By adjusting the geometry and repeating wind tunnel tests, further refining of the design and will potentially **improve the lift-to-drag ratio** even more.

References

Jia, X., Chang, T., Zhou, S., et al. (2025). Progress in bionic deformable wing of aircraft driven by shape memory alloy. Journal of Bionic Engineering. <https://doi.org/10.1007/s42235-025-00744-2>

Reddy, S., Sobieczky, H., Abdoli, A., & Dulikravich, G. (2014). Winglets – Multi-objective optimization of aerodynamic shapes. AIAA. <https://doi.org/10.2514/6.2015-1489>

Wang, W., & Yuan, G. (2024). Review on the structure design of morphing winglets. Aerospace, 11(12), 1004. <https://doi.org/10.3390/aerospace11121004>